CUSTOMER SERVICE ANALYTICS DASHBOARD

Power BI Portfolio Case Study

PROJECT	**Fly Analytics - Energy Company Service Operations Dashboard**
ROLE	IT Analyst
DURATION	1 Month Development | Production Deployment
TOOLS	Power BI Desktop, DAX, Power Query, DAX Studio
CONTEXT	Enterprise analytics solution deployed for operations management
STATUS	Actively used by management for daily decision-making

EXECUTIVE SUMMARY

This case study showcases a production Power BI analytics solution I delivered as an IT Analyst to address critical visibility gaps in customer service operations for an energy company. The dashboard consolidates data across ticketing systems, agent performance, service channels, and process management into a unified 13-page analytical platform currently used daily by management for operational decision-making.

The solution tracks 28,000+ tickets, monitors performance across multiple regions and branches, analyzes 75,000+ process records, and provides granular insights into service levels, wait times, and agent productivity. Deployed to production after one month of development, the platform now serves as the primary tool for staffing decisions, performance reviews, and resource optimization. Key features include SLA compliance tracking, peak hour analysis, month-over-month comparisons, and customer satisfaction metrics.

BUSINESS CHALLENGE

The Problem

The organization faced a critical gap in operational visibility: there was no centralized system to measure and track agent productivity. Customer service managers had no objective way to:

- Monitor individual agent performance against service level agreements (SLAs)
- Identify productivity patterns across different times of day, days of week, or service groups
- Compare performance across regions and branches
- Track process completion times and bottlenecks
- Measure customer satisfaction and service quality metrics
- Make data-driven staffing and resource allocation decisions

This lack of visibility resulted in inefficient resource allocation, inconsistent service quality, and limited ability to identify and reward top performers or coach underperformers.

Solution Approach

As the IT Analyst assigned to this initiative, I developed a comprehensive Power BI analytics platform in one month, following an agile, iterative development process from requirements gathering through production deployment:

- Week 1 - Foundation: Stakeholder interviews, requirements documentation, data discovery, building the data model, and establishing relationships between ticketing, agent, service, and process data sources
- Week 2 - Core Dashboards: Developed primary operational views (Home, Peak Hours, Agents, Groups) with key performance indicators and service level tracking for management review
- Week 3 - Advanced Analytics: Built process analysis pages, SLA monitoring, comparative views for services and groups, and implemented user feedback from initial demos
- Week 4 - Production Readiness: Optimized DAX calculations using DAX Studio, implemented final interactivity features, conducted user acceptance testing, and deployed to production

The solution was designed with two primary user personas: managers who need high-level operational oversight for staffing decisions and performance comparisons, and individual agents who track their own productivity metrics. Post-deployment, the dashboard became the primary tool for daily operations monitoring and eliminated manual reporting processes that previously took days.

TECHNICAL IMPLEMENTATION

Solution Approach

TOOL	PURPOSE	KEY USAGE
Power BI Desktop	Visualization & Reporting	Dashboard design, interactive filters, drill-throughs
DAX	Calculations & Measures	Time intelligence, custom KPIs, conditional logic
Power Query (M	Data Transformation	ETL processes, data cleaning, column derivations
DAX Studio	Performance Optimization	Query optimization, measure testing, performance tuning

Data Model Architecture

The solution integrates four primary data sources into a star schema model:

- **Tickets (Fact Table):** 28,000+ customer service tickets with issue dates, wait times, service times, abandonments, and regional data
- **Agents (Dimension):** 60+ customer service representatives with performance metrics, assignments, and historical data
- **Services (Dimension):** Service catalog including damages, claims, financing, notifications, and general inquiries
- **Processes (Fact Table):** 75,000+ process records tracking claims, complaints, petitions, suggestions, and orders with completion dates and agent assignments

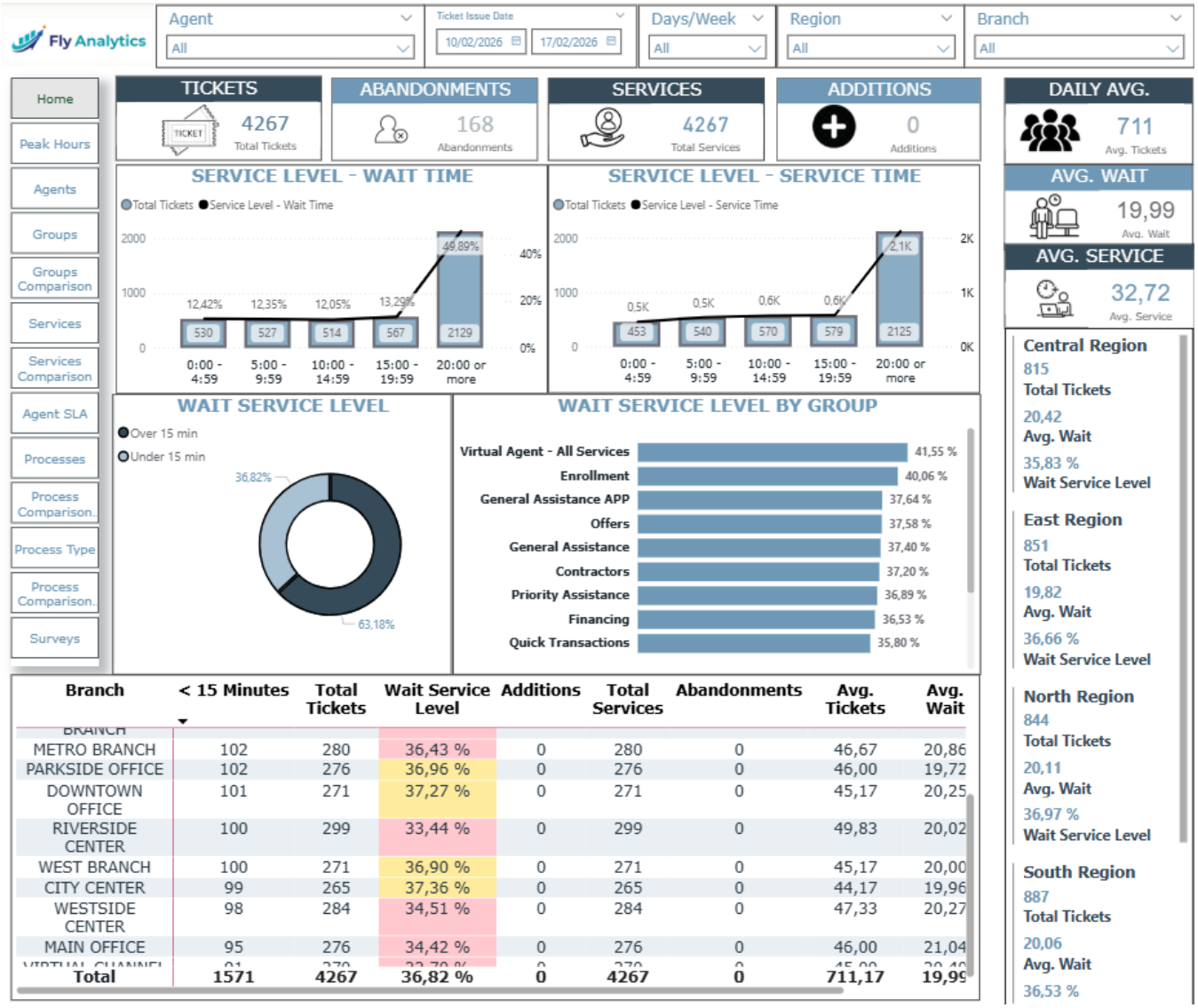
Key Technical Features

- **Custom DAX Measures:** Developed complex time-intelligence calculations including dynamic month comparisons (MesesDisponibles1, MesesDisponibles11) with multi-language support
- **Performance Optimization:** Used DAX Studio to identify and optimize slow-running measures, reducing page load times by analyzing query plans and storage engine activity
- **Dynamic Filtering:** Implemented cascading filters across regions, branches, agents, and time periods with bidirectional cross-filtering
- **Conditional Formatting:** Color-coded performance indicators (green for meeting SLA targets, yellow for borderline, red for underperformance)
- **Drill-Through Actions:** Enabled users to click on summary metrics and navigate to detailed agent-level or transaction-level data
- **Bookmarks & Navigation:** Created intuitive navigation between 13 dashboard pages with consistent layout and branding

DASHBOARD DEEP DIVE

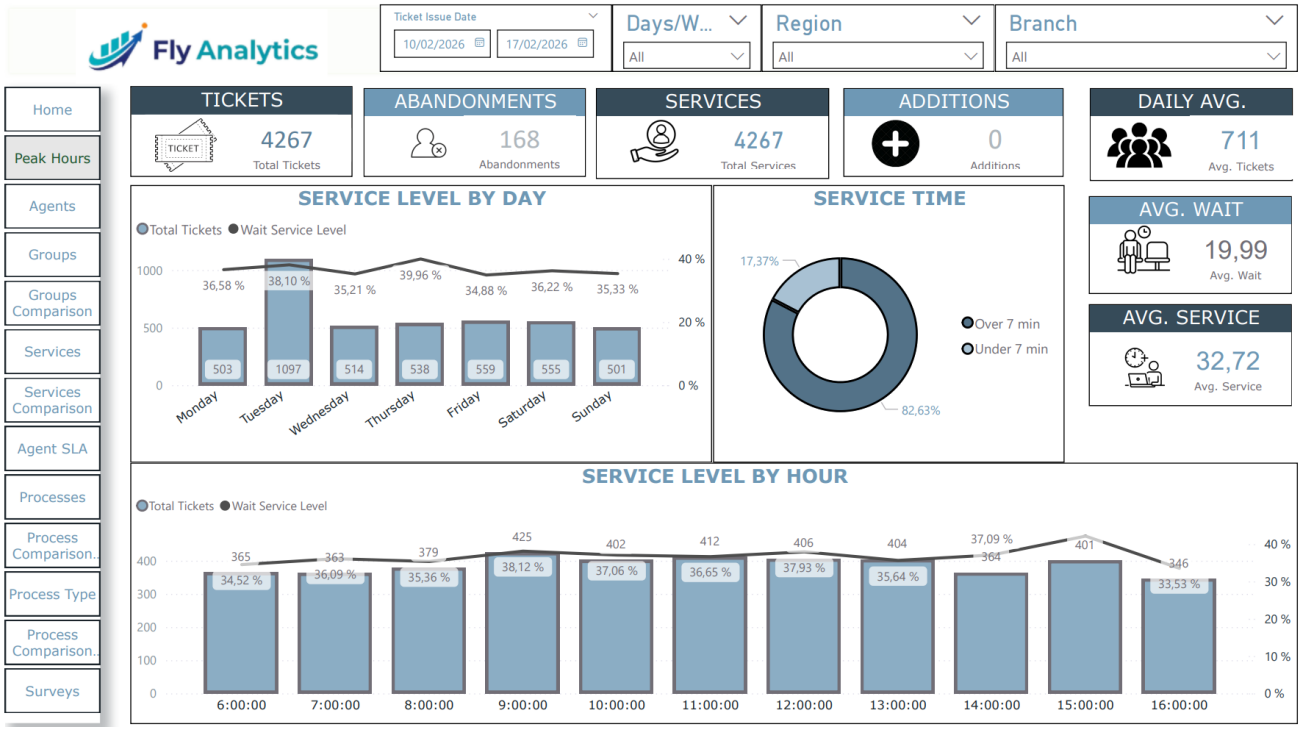
Dashboard Screenshots & Analysis

The dashboard consists of 13 interconnected pages providing comprehensive operational visibility. Below are screenshots of key pages with analysis:

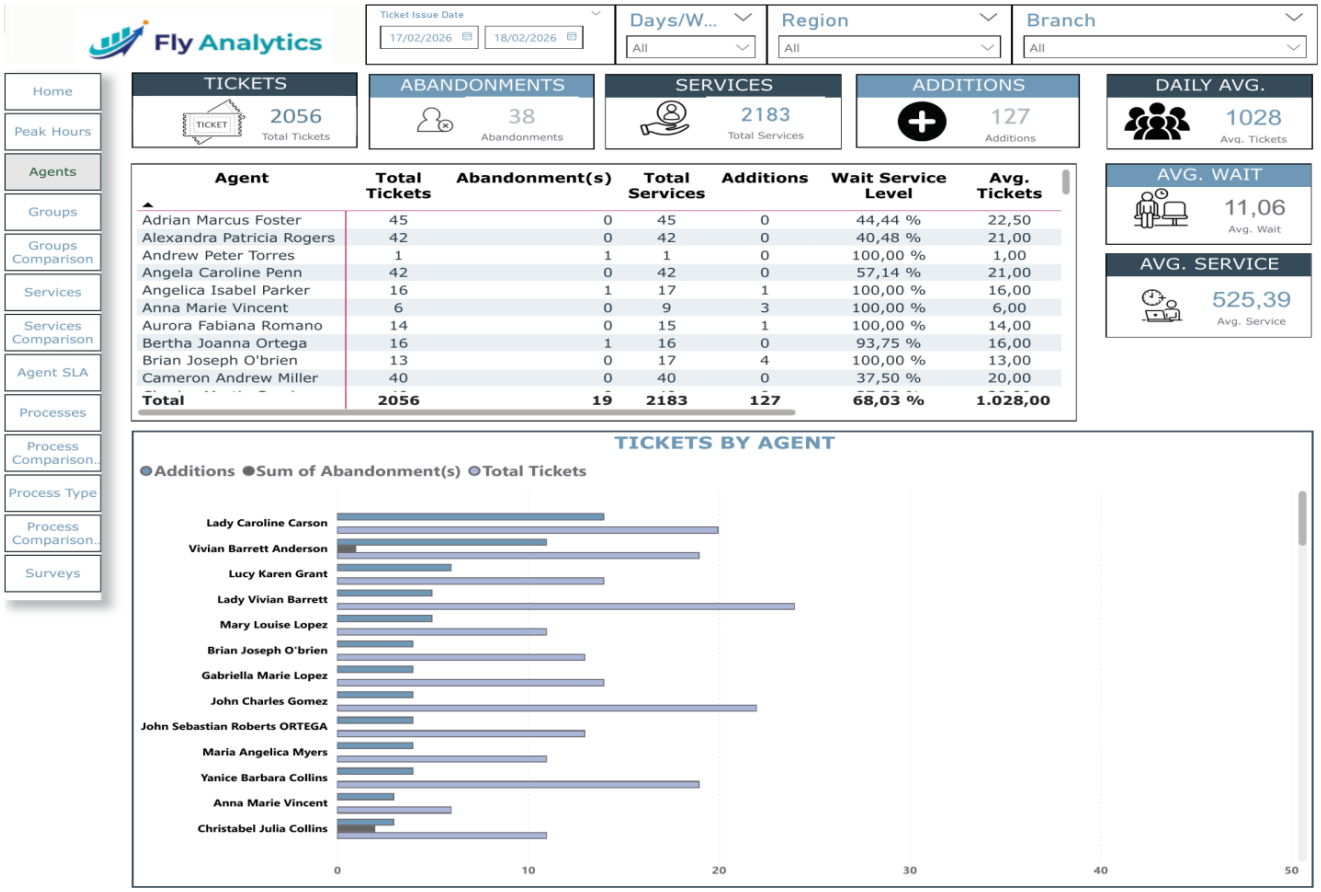


Branch	< 15 Minutes	Total Tickets	Wait Service Level	Additions	Total Services	Abandonments	Avg. Tickets	Avg. Wait
METRO BRANCH	102	280	36,43 %	0	280	0	46,67	20,86
PARKSIDE OFFICE	102	276	36,96 %	0	276	0	46,00	19,72
DOWNTOWN OFFICE	101	271	37,27 %	0	271	0	45,17	20,25
RIVERSIDE CENTER	100	299	33,44 %	0	299	0	49,83	20,02
WEST BRANCH	100	271	36,90 %	0	271	0	45,17	20,00
CITY CENTER	99	265	37,36 %	0	265	0	44,17	19,96
WESTSIDE CENTER	98	284	34,51 %	0	284	0	47,33	20,27
MAIN OFFICE	95	276	34,42 %	0	276	0	46,00	21,04
Total	**1571**	**4267**	**36,82 %**	**0**	**4267**	**0**	**711,17**	**19,99**

Home Dashboard: Executive overview showing 4,267 tickets, 36.82% service level compliance, regional performance breakdown, and wait time distribution. Donut charts and bar graphs provide at-a-glance KPI monitoring.

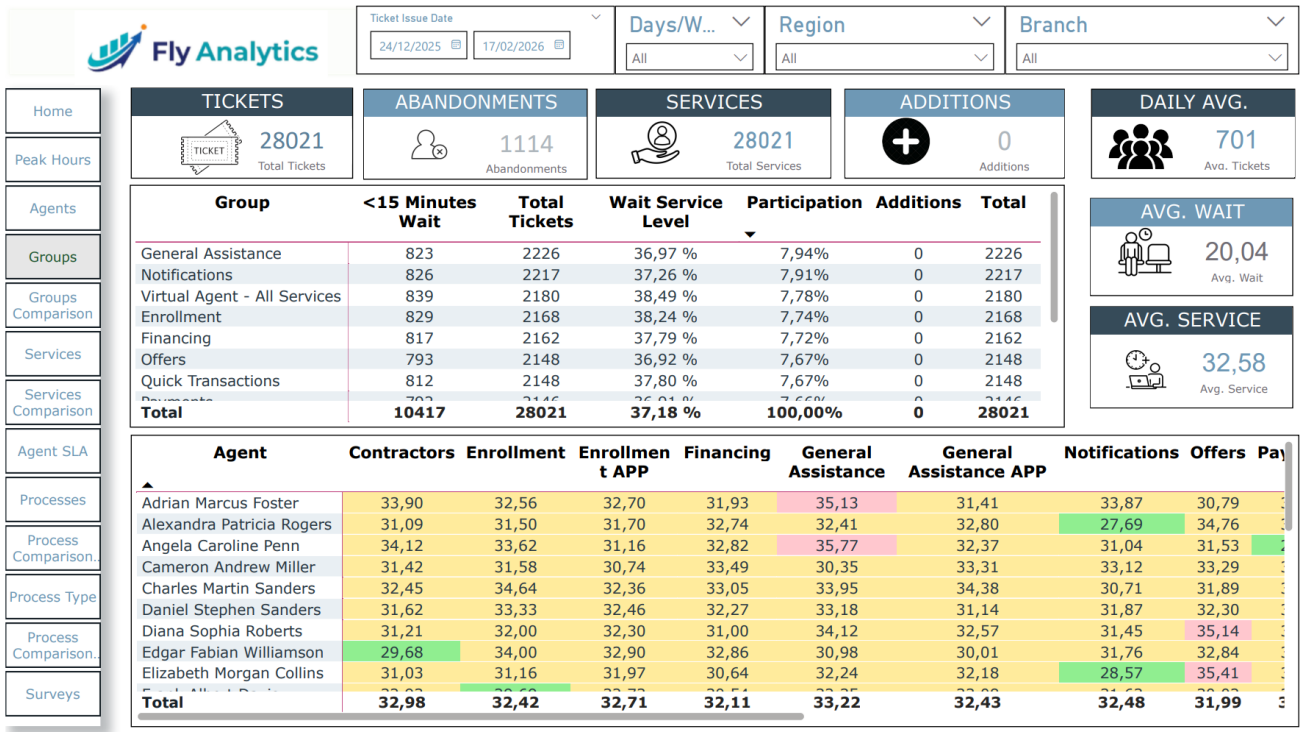


Peak Hours Analysis: Hourly and daily service patterns reveal optimal staffing periods. Tuesday peaks at 1,097 tickets, while 9:00 AM shows highest hourly volume (425 tickets, 38.12% SLA).



Agent	Total Tickets	Abandonment(s)	Total Services	Additions	Wait Service Level	Avg. Tickets
Adrian Marcus Foster	45	0	45	0	44,44 %	22,50
Alexandra Patricia Rogers	42	0	42	0	40,48 %	21,00
Andrew Peter Torres	1	1	1	0	100,00 %	1,00
Angela Caroline Penn	42	0	42	0	57,14 %	21,00
Angelica Isabel Parker	16	1	17	1	100,00 %	16,00
Anna Marie Vincent	6	0	9	3	100,00 %	6,00
Aurora Fabiana Romano	14	0	15	1	100,00 %	14,00
Bertha Joanna Ortega	16	1	16	0	93,75 %	16,00
Brian Joseph O'brien	13	0	17	4	100,00 %	13,00
Cameron Andrew Miller	40	0	40	0	37,50 %	20,00
Total	**2056**	**19**	**2183**	**127**	**68,03 %**	**1.028,00**

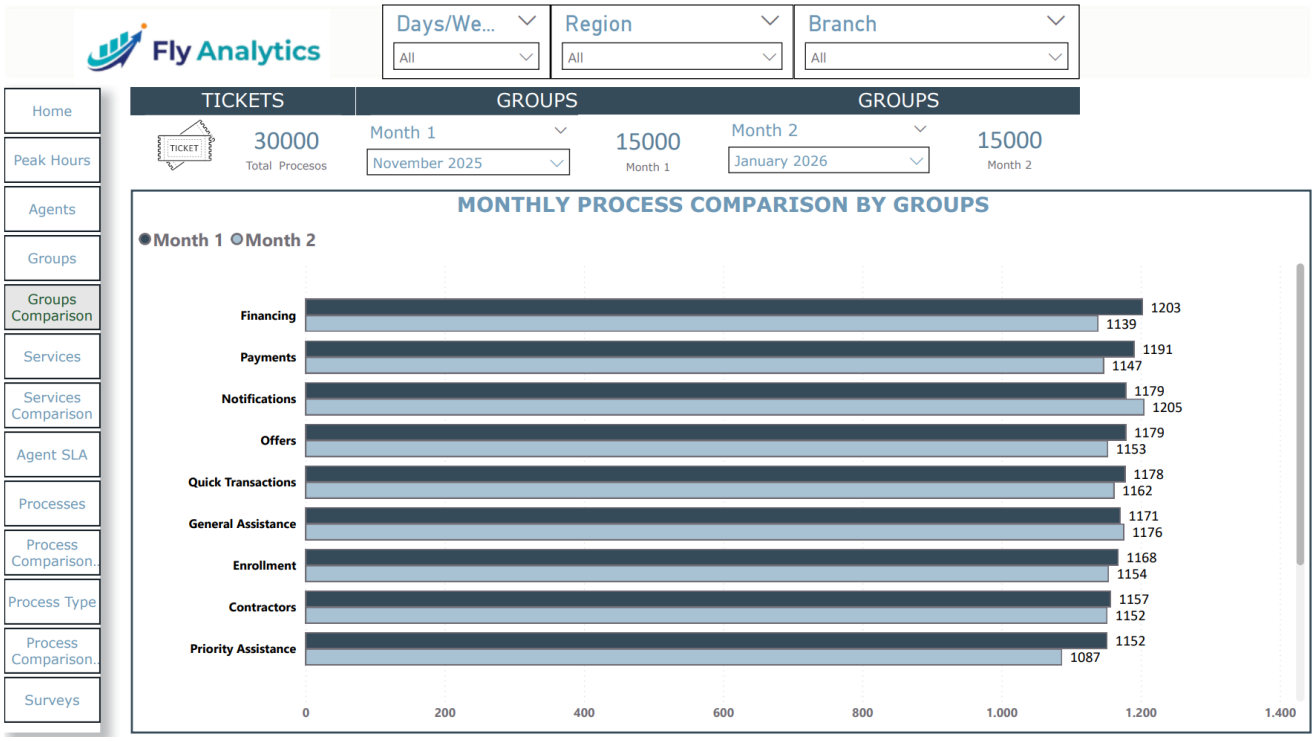
Agent Performance: Individual agent tracking with 2,056 tickets and 68.03% wait service level. Bar chart rankings identify top performers for recognition and coaching opportunities.



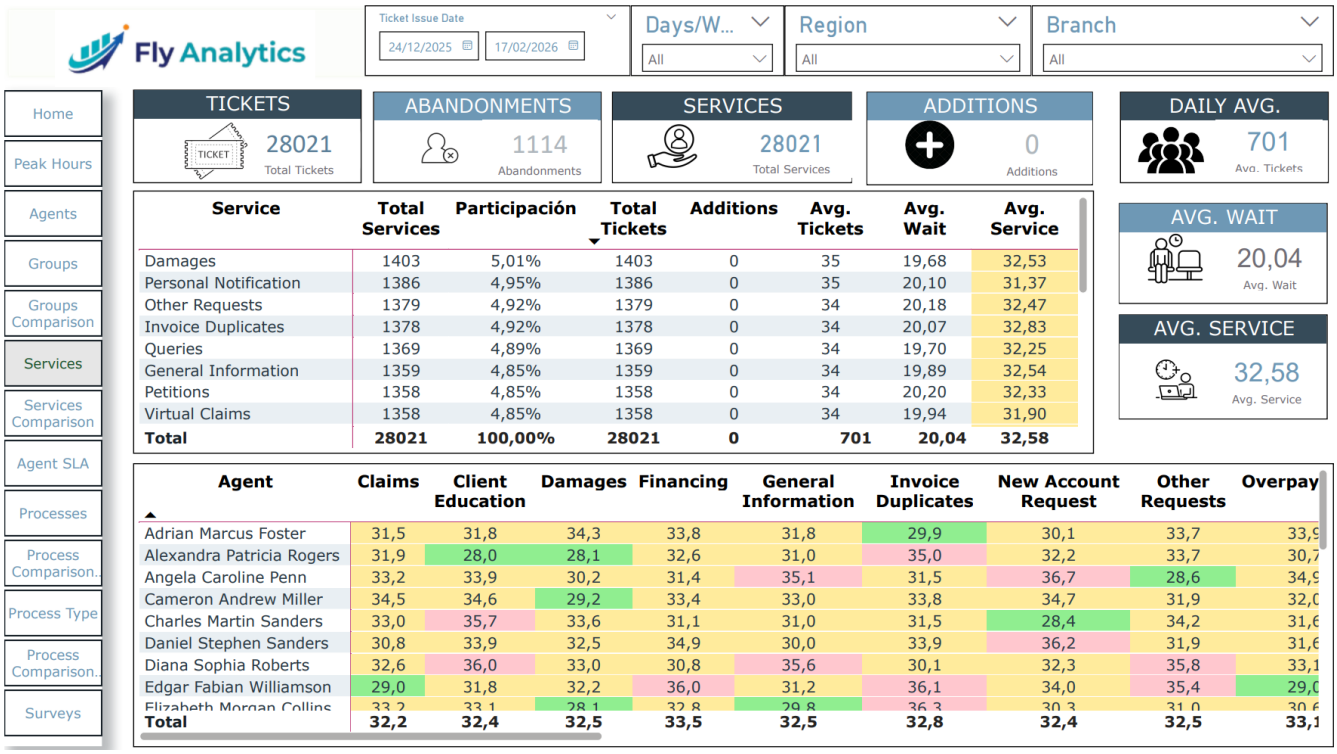
Group	<15 Minutes Wait	Total Tickets	Wait Service Level	Participation	Additions	Total
General Assistance	823	2226	36,97 %	7,94%	0	2226
Notifications	826	2217	37,26 %	7,91%	0	2217
Virtual Agent - All Services	839	2180	38,49 %	7,78%	0	2180
Enrollment	829	2168	38,24 %	7,74%	0	2168
Financing	817	2162	37,79 %	7,72%	0	2162
Offers	793	2148	36,92 %	7,67%	0	2148
Quick Transactions	812	2148	37,80 %	7,67%	0	2148
Total	**10417**	**28021**	**37,18 %**	**100,00%**	**0**	**28021**

Agent	Contractors	Enrollment	Enrollment APP	Financing	General Assistance	General Assistance APP	Notifications	Offers	Pa
Adrian Marcus Foster	33,90	32,56	32,70	31,93	35,13	31,41	33,87	30,79	
Alexandra Patricia Rogers	31,09	31,50	31,70	32,74	32,41	32,80	27,69	34,76	
Angela Caroline Penn	34,12	33,62	31,16	32,82	35,77	32,37	31,04	31,53	
Cameron Andrew Miller	31,42	31,58	30,74	33,49	30,35	33,31	33,12	33,29	
Charles Martin Sanders	32,45	34,64	32,36	33,05	33,95	34,38	30,71	31,89	
Daniel Stephen Sanders	31,62	33,33	32,46	32,27	33,18	31,14	31,87	32,30	
Diana Sophia Roberts	31,21	32,00	32,30	31,00	34,12	32,57	31,45	35,14	
Edgar Fabian Williamson	29,68	34,00	32,90	32,86	30,98	30,01	31,76	32,84	
Elizabeth Morgan Collins	31,03	31,16	31,97	30,64	32,24	32,18	28,57	35,41	
Total	**32,98**	**32,42**	**32,71**	**32,11**	**33,22**	**32,43**	**32,48**	**31,99**	

Groups: Service group analysis with participation percentages and agent performance matrix across different service types with conditional formatting.



Groups Comparison: Month-over-month trends (Nov 2025 vs Jan 2026) showing Financing leads with 1,203→1,139 processes.

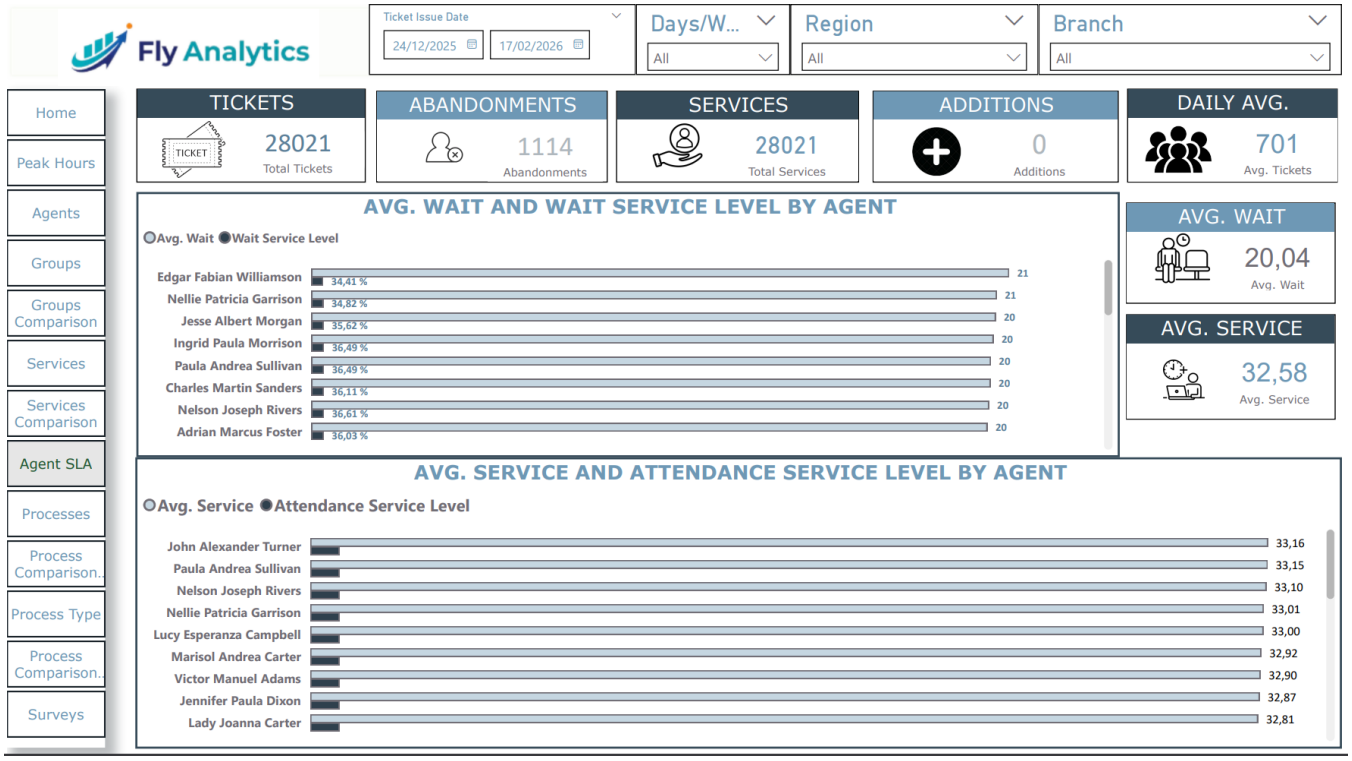


Service	Total Services	Participación	Total Tickets	Additions	Avg. Tickets	Avg. Wait	Avg. Service
Damages	1403	5,01%	1403	0	35	19,68	32,53
Personal Notification	1386	4,95%	1386	0	35	20,10	31,37
Other Requests	1379	4,92%	1379	0	34	20,18	32,47
Invoice Duplicates	1378	4,92%	1378	0	34	20,07	32,83
Queries	1369	4,89%	1369	0	34	19,70	32,25
General Information	1359	4,85%	1359	0	34	19,89	32,54
Petitions	1358	4,85%	1358	0	34	20,20	32,33
Virtual Claims	1358	4,85%	1358	0	34	19,94	31,90
Total	**28021**	**100,00%**	**28021**	**0**	**701**	**20,04**	**32,58**

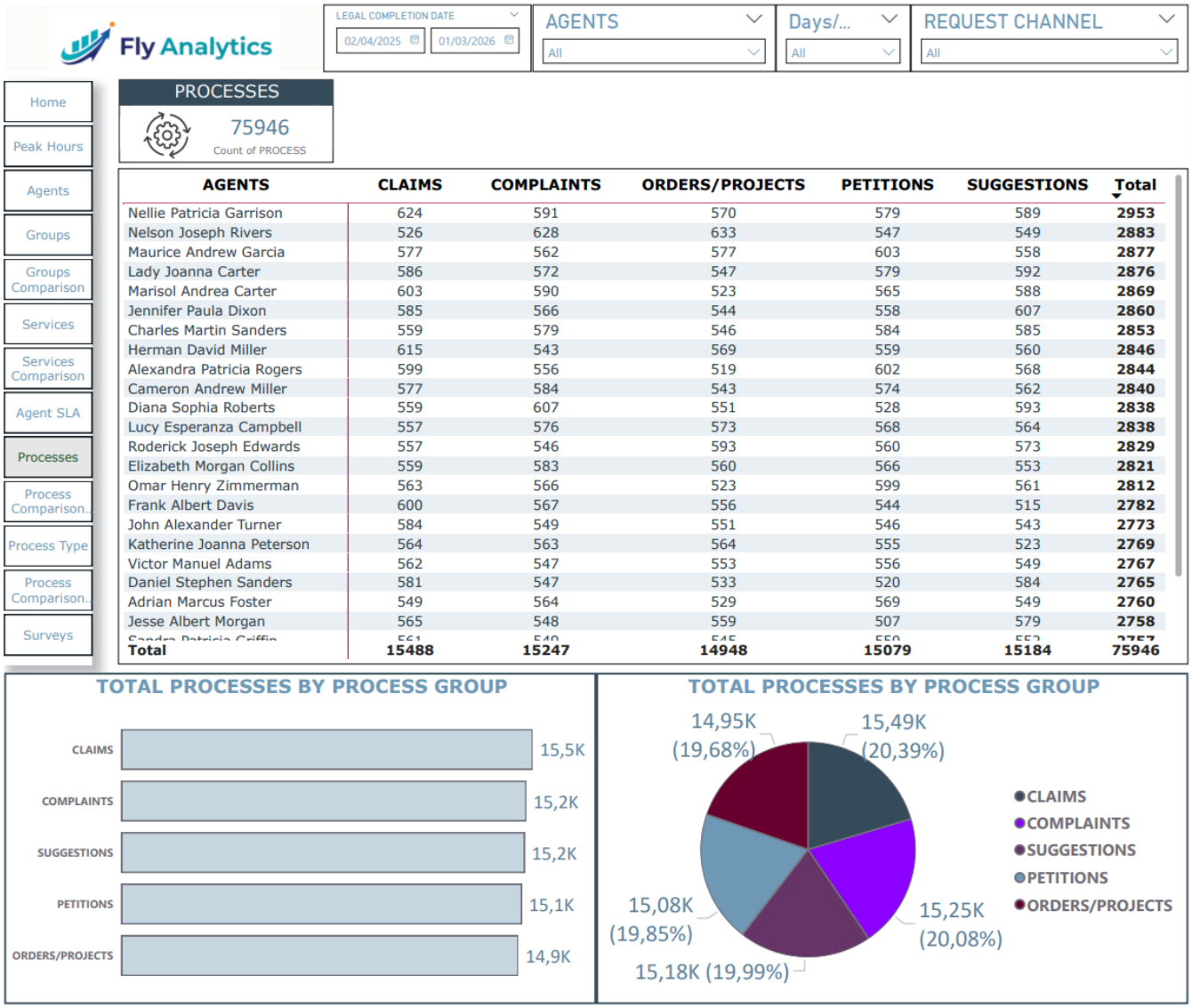
Agent	Claims	Client Education	Damages	Financing	General Information	Invoice Duplicates	New Account Request	Other Requests	Overpay
Adrian Marcus Foster	31,5	31,8	34,3	33,8	31,8	29,9	30,1	33,7	33,9
Alexandra Patricia Rogers	31,9	28,0	28,1	32,6	31,0	35,0	32,2	33,7	30,7
Angela Caroline Penn	33,2	33,9	30,2	31,4	35,1	31,5	36,7	28,6	34,9
Cameron Andrew Miller	34,5	34,6	29,2	33,4	33,0	33,8	34,7	31,9	32,0
Charles Martin Sanders	33,0	35,7	33,6	31,1	31,0	31,5	28,4	34,2	31,6
Daniel Stephen Sanders	30,8	33,9	32,5	34,9	30,0	33,9	36,2	31,9	31,6
Diana Sophia Roberts	32,6	36,0	33,0	30,8	35,6	30,1	32,3	35,8	33,1
Edgar Fabian Williamson	29,0	31,8	32,2	36,0	31,2	36,1	34,0	35,4	29,0
Elizabeth Morgan Collins	33,2	33,1	28,1	32,8	29,8	36,3	30,3	31,0	30,6
Total	**32,2**	**32,4**	**32,5**	**33,5**	**32,5**	**32,8**	**32,4**	**32,5**	**33,1**

Services: Detailed service-level metrics across 28,021 tickets. Top services: Damages (5.01%), Personal Notification (4.95%), Other Requests (4.92%).

Services Comparison: Temporal analysis (Jan 2026 vs Oct 2025) identifying seasonal patterns. Petitions increased 757→778, Invoice Duplicates 696→753.

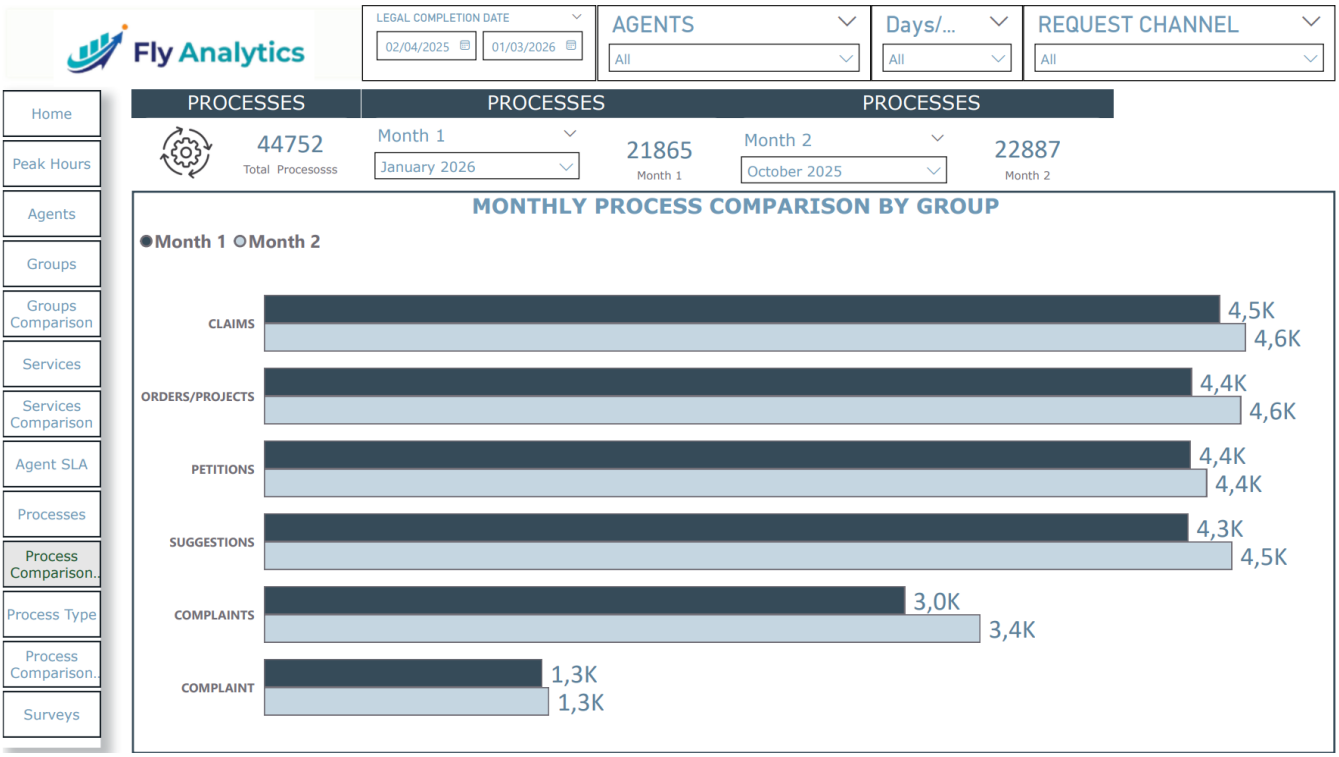


Agent SLA: Dual-metric tracking (wait time & service time) by agent. Performance spread: Edgar Williamson 34.41% vs Angela Penn 57.14% service level.



AGENTS	CLAIMS	COMPLAINTS	ORDERS/PROJECTS	PETITIONS	SUGGESTIONS	Total
Nellie Patricia Garrison	624	591	570	579	589	**2953**
Nelson Joseph Rivers	526	628	633	547	549	**2883**
Maurice Andrew Garcia	577	562	577	603	558	**2877**
Lady Joanna Carter	586	572	547	579	592	**2876**
Marisol Andrea Carter	603	590	523	565	588	**2869**
Jennifer Paula Dixon	585	566	544	558	607	**2860**
Charles Martin Sanders	559	579	546	584	585	**2853**
Herman David Miller	615	543	569	559	560	**2846**
Alexandra Patricia Rogers	599	556	519	602	568	**2844**
Cameron Andrew Miller	577	584	543	574	562	**2840**
Diana Sophia Roberts	559	607	551	528	593	**2838**
Lucy Esperanza Campbell	557	576	573	568	564	**2838**
Roderick Joseph Edwards	557	546	593	560	573	**2829**
Elizabeth Morgan Collins	559	583	560	566	553	**2821**
Omar Henry Zimmerman	563	566	523	599	561	**2812**
Frank Albert Davis	600	567	556	544	515	**2782**
John Alexander Turner	584	549	551	546	543	**2773**
Katherine Joanna Peterson	564	563	564	555	523	**2769**
Victor Manuel Adams	562	547	553	556	549	**2767**
Daniel Stephen Sanders	581	547	533	520	584	**2765**
Adrian Marcus Foster	549	564	529	569	549	**2760**
Jesse Albert Morgan	565	548	559	507	579	**2758**
Sandra Patricia Griffin	561	540	545	550	552	**2757**
Total	**15488**	**15247**	**14948**	**15079**	**15184**	**75946**

Processes: 75,946 process records across Claims (20.39%), Complaints (20.08%), Suggestions (19.99%), Petitions (19.85%), Orders/Projects (19.68%).



Process Comparison (Groups): Monthly trends showing Claims 4.5K→4.6K, Complaints 3.0K→3.4K growth between periods.

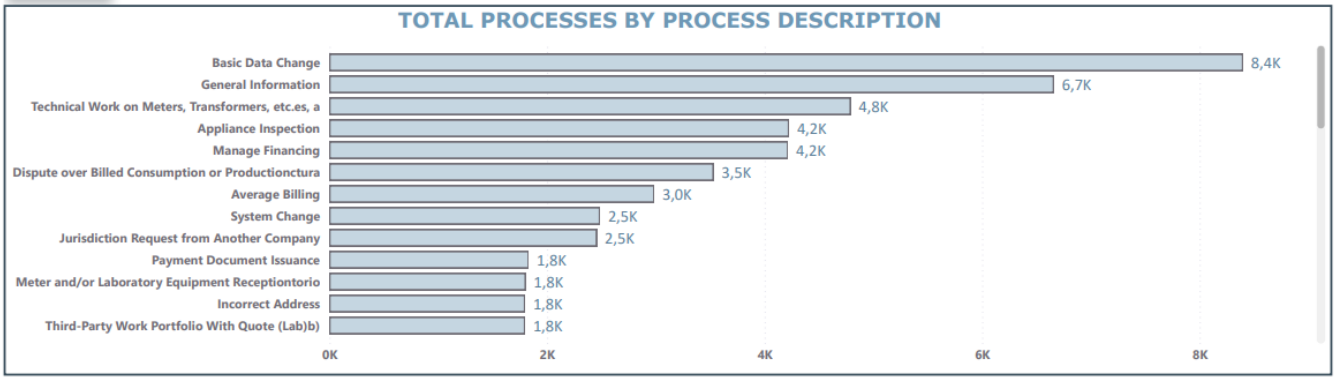
Fly Analytics

COMPLETION DATE 02/04/2025 01/03/2026 | AGENTS All | Days/... All | REQUEST CHANNEL All

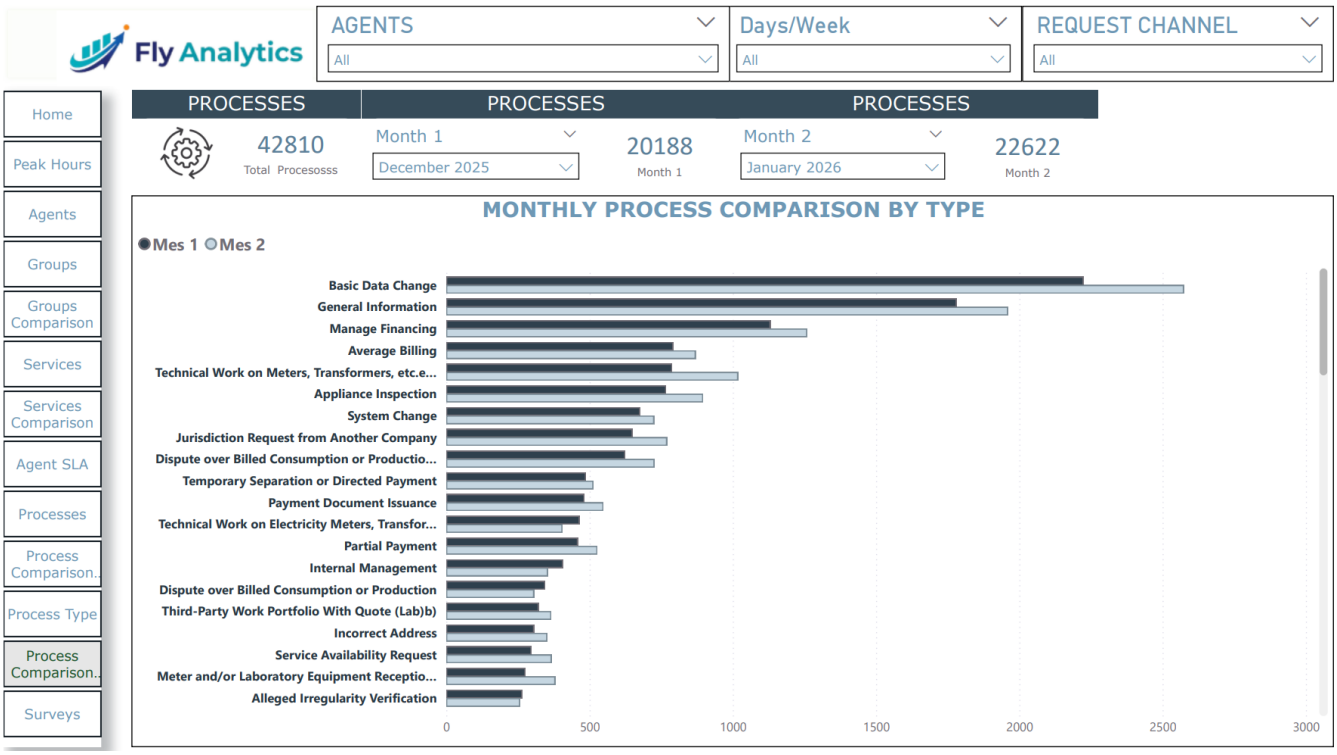
Home
Peak Hours
Agents
Groups
Groups Comparison
Services
Services Comparison
Agent SLA
Processes
Process Comparison..
Process Type
Process Comparison..
Surveys

PROCESSES
75946
Count of PROCESS

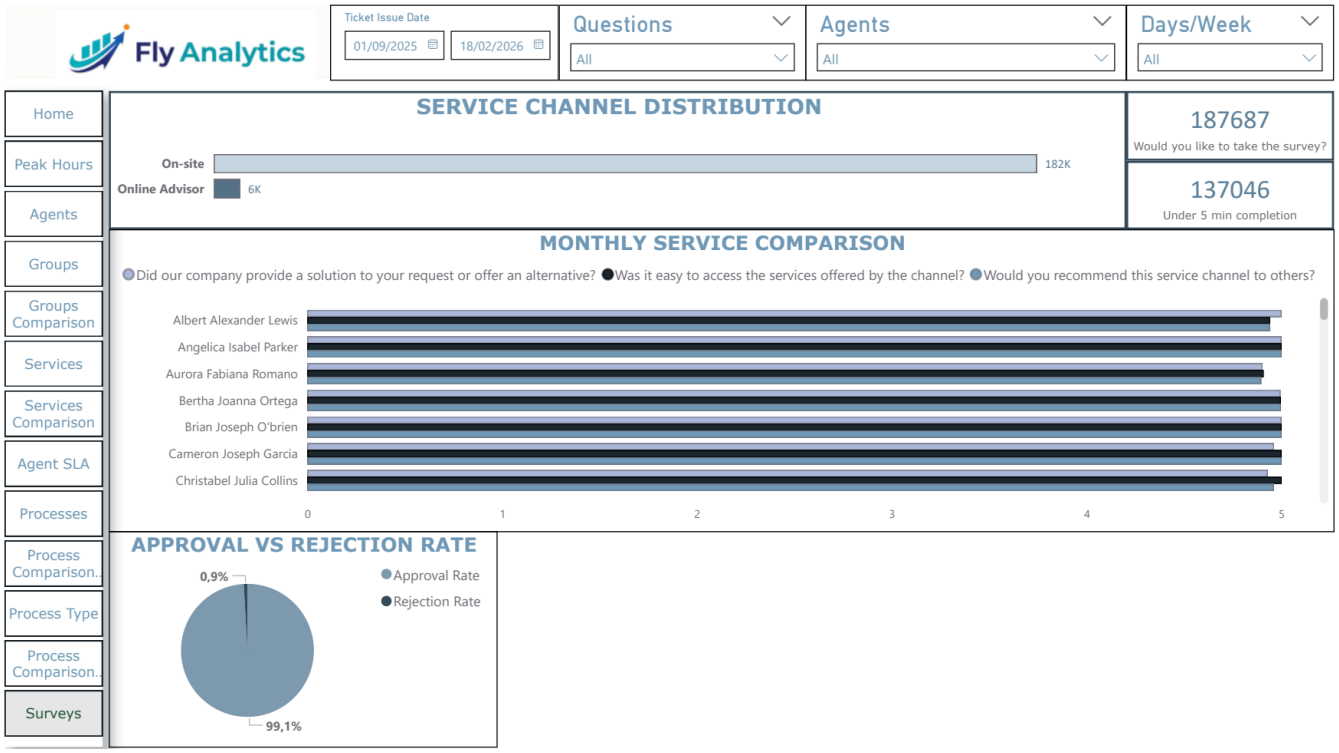
AGENTS	Account Activation	Adjustments	Alleged Irregularity Verification	APD AGE Data Update	Appliance Inspection	Average Billing	Basic Data Change	Comr Re
Adrian Marcus Foster	45	47	53	19	141	107	316	
Alexandra Patricia Rogers	57	36	48	24	174	108	291	
Angela Caroline Penn	38	39	40	23	146	121	294	
Cameron Andrew Miller	47	38	50	16	154	117	331	
Charles Martin Sanders	43	46	46	23	177	90	311	
Daniel Stephen Sanders	53	37	41	14	163	116	306	
Diana Sophia Roberts	44	39	44	26	172	117	318	
Edgar Fabian Williamson	32	45	46	20	152	109	297	
Elizabeth Morgan Collins	56	51	39	18	173	78	324	
Frank Albert Davis	44	39	41	17	152	120	332	
Herman David Miller	47	51	54	21	152	119	330	
Ingrid Paula Morrison	39	45	42	22	143	107	311	
Jennifer Paula Dixon	54	39	40	25	172	101	325	
Jesse Albert Morgan	47	36	41	21	168	120	304	
John Alexander Turner	44	49	40	21	144	116	285	
Katherine Joanna Peterson	55	50	56	20	183	118	309	
Lady Joanna Carter	49	53	44	16	139	126	315	
Lucy Esperanza Campbell	45	61	43	29	166	125	318	
Marisol Andrea Carter	47	42	41	22	162	113	310	
Maurice Andrew Garcia	53	36	39	14	142	107	327	
Nellie Patricia Garrison	47	51	58	31	149	126	319	
Total	**1276**	**1206**	**1221**	**582**	**4222**	**2983**	**8396**	



Process Type: Granular analysis: Basic Data Change (8.4K), General Information (6.7K), Technical Work (4.8K) with agent contribution matrix.



Process Comparison (Type): Month-over-month (Dec 2025: 20,188 vs Jan 2026: 22,622) for resource planning and forecasting.



Surveys: 187,687 survey responses, 137,046 under 5 min completion. 99.1% approval rate. Channel split: On-site (182K) vs Online Advisor (6K).